Project: Causal Machine Learning

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1 Introduction

We consider estimation of the effect of eligibility and participation in 401(k) (this is a tax-advantaged retirement savings plan) on accumulated assets using 9 915 observations at the household level, drawn from the Survey of Income and Program Participation:

```
library(hdm)
data(pension)
```

In this project, each data analysis question must be based on debiased/targeted machine learning (unless otherwise stated). Make sure to use 5- or 10-fold cross-fitting in all such analyses, and to base these either on a library of complementary algorithms, or a single, but carefully tuned algorithm. In all data analyses, you are expected to report an estimate and 95% confidence interval, if possible, and to interpret the results in an accessible manner. In addition, you are expected to make some checks to evaluate and report the quality of the results (e.g., you may want to check the stability of inverse probability weights, or whether some individuals have extreme influence functions), even when this is not explicitly asked.

2 Task

The key problem in determining the effect of participation in 401(k) plans on accumulated assets is the fact that the decision to enroll in a 401(k) is non-random. It is generally recognized that some people have a higher preference for saving than others. It also seems likely that those individuals with high unobserved preference for saving would be most likely to choose to participate in tax-advantaged retirement savings plans and would tend to have otherwise high amounts of accumulated assets. The presence of unobserved savings preferences with these properties then implies that conventional estimates that do not account for confounding will be biased upward, tending to overstate the savings effects of 401(k) participation.

We will first focus on **the effect of eligibility for enrolling in a 401(k) plan**. Around the time 401(k)'s initially became available, people were unlikely to be basing their employment decisions on whether an employer offered a 401(k) but would instead focus on income. Thus, eligibility for a 401(k) could be taken as exchangeable conditional on income, and the causal effect of 401(k) eligibility could be directly estimated by comparing eligible and ineligible individuals with the same income.

1. Use causal forests to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets, in function of age, income, family size, years of education, a married indicator, a two-earner status indicator, a defined benefit pension status indicator, an IRA (individual retirement account) participation indicator, and a home ownership indicator.

   (a) Provide insight into the results (e.g., think of an informative visual).

   (b) It was previously stated that eligibility for a 401(k) could be taken as exchangeable conditional on income. Is it problem that the analysis additionally conditions on other variables, e.g. age, ...? Why (not)? You may consider the use of a causal diagram to support your reasoning.

2. Use `ntile` to create 4 subgroups based on the obtained predictions from the causal forest and estimate the average effect of eligibility for enrolling in a 401(k) plan on net financial assets in each subgroup, along with 95% confidence intervals.

3. Assess if there is evidence of effect heterogeneity by estimating

$$\sigma^2 = \text{Var}\left\{E(Y^1 - Y^0|L)\right\},$$

   with L the variables used in the causal forest. For this, use that the efficient influence function of σ^2 is

$$\begin{aligned}&\left\{E(Y^1 - Y^0|L) - E(Y^1 - Y^0)\right\}^2 - \sigma^2\\&+2\left\{E(Y^1 - Y^0|L) - E(Y^1 - Y^0)\right\}\left\{\frac{A}{P(A=1|L)} - \frac{1-A}{P(A=0|L)}\right\}\\&\times\{Y - E(Y|A,L)\}\end{aligned}$$

   Explain how you performed the analysis, report an estimate of σ^2 and corresponding 95% confidence interval, and interpret the results.

4. The causal forests that were used before have the drawback that they learn the causal effect conditional on many variables, making it both difficult to learn with good accuracy and difficult to visualize. Instead, we will therefore estimate the average effect of eligibility for enrolling in a 401(k) plan on net financial assets *conditional on only* income. For this, use an orthogonal learner, but control for the same variables as before (not just income, but also the other variables in order to have better guarantees of a valid confounding adjustment). Note that the causal forests function in the `grf` package may not allow you to do that, and that you may need to use a different orthogonal learner as discussed in class.

(a) Report a clear visual of the results, displaying the estimated average effect of eligibility for enrolling in a 401(k) plan in function of income.

(b) Clarify if the chosen learner makes specific assumptions that were not made by the earlier causal forest.

Make a **concise, but complete** report describing what analyses you performed and what are your conclusions to the questions. There is no need to be extensive in your report. For instance, there is no need to repeat the theory, as you may assume that the reader is familiar with the procedures that you used. However, you should carefully interpret the reported estimates and make sure that their meaning is clear and not obscure. Overall, I expect that 3 pages should suffice; 5 pages is a maximum.

You can work on this project in groups of 2 or 3 people. It is recommended that you split the work, **and then give feedback on each other's report**. If you split the work, the project should not take too much work, but otherwise it may. One report per group must be returned no later than no later than May 15, together with structured software code (i.e. a script file in R or Python). All the files that you submit, must carry the family name of one of the members in the group.

This project is a group assignment, but your grade will reflect your individual contribution. At the end of the project, each group member must report confidentially and individually (via the Comments box when you submit the assignment on Ufora) what is your assessment of the percentage contribution of each group member, including yourself. These percentages must sum to 100%. Your individual grade will be scaled according to the average contribution score received from all group members, including yourself. If you rate a group member to have contributed less than half of an equal share by all other group members, then please add a brief explanation to your email. A student who is unanimously rated as having contributed less than half of an equal share by all other group members, and who cannot demonstrate sufficient understanding of the group's work in a follow-up conversation with the instructor, will receive a grade of zero for this project, regardless of the quality of the group's submission.

If personal circumstances prevent you from contributing fully to the group project, please notify the instructor as soon as possible. Your peers should not bear the consequences of your absence, and early communication gives the group time to adjust. Claiming a fair share of the grade without having made a fair contribution is a serious form of academic dishonesty that will be regarded as fraud. Students found to have done so will receive a grade of zero for this project and may be subject to further disciplinary action in accordance with university policy.

Finally, resist the temptation to have AI tools complete the project for you. Doing so means depriving yourself of the core learning experience this assignment is designed to provide. Understanding develops through thinking, trying, making mistakes, and working through them – not through copying output you did not generate yourself. A project submitted with little genuine intellectual effort may look polished, but it will leave you without the understanding that this course is meant to build, and that you will be expected to demonstrate in the exam.